

理想与现实：中国年轻女性学习编程（“转码”）成因探究

硕士毕业论文

成绩：A-（专业排名第二，前10%）

摘要

1. 研究背景与问题

随着中国互联网行业的飞速发展，编程能力为求职者提供了更广阔的职业选择。近年来，在中国出现了一个显著的趋势：越来越多非计算机专业的“90后”和“00后”女性选择学习编程（这一现象在中文互联网被称为“转码”），并主动进入由男性主导的IT行业。以往关于STEM（科学、技术、工程和数学）领域女性的研究，大多聚焦于解释女性为何在高中或大学后“离开”STEM领域（即“漏水管”现象）。然而，针对中国年轻女性为何在面临男性主导的职场环境下，依然主动跨专业“进入”该领域的动机与现实困境，现有文献缺乏充分的探讨。本研究旨在填补这一空白，深入探究中国女性选择“转码”的原因、学习过程中的主要方式，以及她们在IT行业不同人生阶段所面临的性别阻力。

2. 研究方法

本研究采用了混合定性研究方法，结合了网络内容分析与深度访谈，以全面捕捉“转码”女性的群体特征与深层心理机制。

内容分析法：研究选取了中国高学历年轻群体最常用的两个社交媒体平台——豆瓣（女性用户占62%）和知乎（男性用户占70%）作为数据来源。研究者筛选了截至2022年12月在这两个平台上点赞数超过200的“转码”相关热门帖子，共计132篇（豆瓣55篇，知乎77篇）。在数据处理上，首先使用Python对帖子正文进行词频统计以提取核心议题，随后对具有代表性的高赞帖子及其评论区进行了深入的定性文本分析。

深度访谈法：为了进一步验证和解释社交媒体上的文本现象，研究者主要通过哈尔滨工业大学的网络论坛招募了12名受访者，并在2023年3月至5月期间通过ZOOM和腾讯会议进行了平均时长为1小时的在线深度访谈。受访者被策略性地分为三组：

A组（5名女性）：本科原专业中包含编程相关课程的理工科女性；B组（5名女性）：本科原专业中无编程课程的女性，其中包含3名典型的高中文科生（纯人文社科背景）；C组（2名男性）：在访谈推进过程中额外增加的男性程序员/计算机博士，旨在从男性同行的视角了解针对女性的差别对待是否出于主动选择。

3. 研究发现

研究揭示了女性在“转码”生命周期的三个不同阶段所面临的动机、行为与现实境遇：

转码动机：教育体制的限制与对经济独立的渴望 女性选择转码的初始原因深受中国教育体制与传统性别观念的影响。高中时期严格的“文理分科”制度、缺乏职业规划的应试教育，以及父母与教师认为“文科更适合女孩”的传统性别偏见，导致许多女性在进入大学前被排除在IT及核心STEM专业之外。在大学阶段意识到原专业的性别偏见或薪资天花板后，她们选择学习编程。原因在于编程主要依赖脑力劳动，女性没有生理劣势，且IT行业能提供更高的薪水和更广阔的就业机会，从而帮助她们实现经济独立与个人自由。

学习过程：自学、留学与“善意性别歧视”的阻碍 由于中国大陆严格的跨专业考研限制，女性很难在国内直接转换为计算机相关的硕士专业。因此，“转码”主要依靠网络自学和出国留学来完成。在学习过程中，她们遇到最大的技术困难与性别无关，而是源于缺乏基础。然而，在校园的小组合作或科研中，她们经常遭遇男性的“善意性别歧视”。男性同学往往以“照顾女性情绪”为由，分配给女性

缺乏技术含量的重复性工作，这种隐形的排斥实质上剥夺了女性接受核心技术训练的机会。值得庆幸的是，豆瓣等平台上互助互勉的女性社群氛围，为她们度过学习难关提供了强大的情感支撑。

职场现实：男性凝视、边缘化与高薪的补偿 进入IT职场后，女性面临着比校园更严峻的“不友好环境（chilly climate）”。社交媒体（如知乎）上充斥着对女性程序员外貌的过度关注与男性凝视。同时，部分男性认为女性在IT求职中享有“性别红利”的观点也与事实不符。实际上，中国互联网企业盛行的“996加班文化”以及男性主导的“兄弟会”社交圈层，使得女性在晋升和融入团队时遭到严重的边缘化。不过，IT行业相对优渥的薪酬和体面的工作机会，在一定程度上缓解了这些性别挑战带来的负面影响。

4. 研究结论与政策建议

尽管IT行业存在着对女性不友好的冷环境，但编程能力作为实现更高收入和职业自主性的工具，促使中国女性不断突破重围主动“转码”。基于上述发现，本研究建议中国的高中应增加职业认知课程，以避免学生因性别刻板印象盲目选科；同时，高校应适当放宽本科生转专业的限制，并为缺乏专业背景的学生提供更多包容性的研究生项目。

关键词： 互联网行业；中国女性；编程；女工程师；性别偏见

Ideals vs. Realities? A Study on Why Young Chinese Women Want to Learn Programming

Master's Dissertation

Grade: A- (Ranked 2nd in the major, Top 10%)

Abstract

1. Research Background and Questions

With the rapid development of China's Internet industry, programming skills provide job seekers with broader career options. In recent years, a significant trend has emerged in China: an increasing number of post-90s and post-00s women from non-computer majors are choosing to learn programming (a phenomenon known as "switch code" on the Chinese Internet) and proactively entering the male-dominated IT industry. Previous studies on women in STEM (Science, Technology, Engineering, and Mathematics) fields have mostly focused on explaining why women "leave" STEM fields after high school or college (i.e., the "leaky pipeline" phenomenon). However, existing literature lacks sufficient discussion regarding the motivations and real-life dilemmas of young Chinese women who proactively cross disciplinary boundaries to "enter" this field despite facing a male-dominated workplace. This study aims to fill this gap by deeply exploring the reasons why Chinese women choose to "switch code," their primary learning methods, and the gender resistance they face at different life stages within the IT industry.

2. Research Methods

This study adopts a mixed qualitative research method, combining online content analysis and in-depth interviews, to comprehensively capture the group characteristics and deep psychological mechanisms of "switch-code" women.

Content Analysis: The study selected Douban (62% female users) and Zhihu (70% male users)—the two most frequently used social media platforms among highly educated young adults in China—as data sources. The researcher screened a total of 132 popular posts (55 from Douban, 77 from Zhihu) related to "switch code" that had received over 200 likes as of December 2022. For data processing, Python was first used to conduct word frequency statistics on the main texts of the posts to extract core topics. Subsequently, an in-depth qualitative textual analysis was performed on representative highly-liked posts and their comment sections.

In-depth Interviews: To further verify and explain the textual phenomena observed on social media, the researcher recruited 12 interviewees, primarily through the online forum of the Harbin Institute of Technology. Online in-depth interviews, averaging one hour in duration, were conducted via ZOOM and Tencent Meeting between March and May 2023. The interviewees were strategically divided into three groups: Group A (5 women): women with STEM backgrounds whose original undergraduate majors included programming-related courses; Group B (5 women): women whose original undergraduate majors had no programming courses, including 3 typical high school liberal arts students (with pure humanities and social science backgrounds); Group C (2 men): male programmers or

computer science PhD students added during the interview process to understand, from the perspective of male peers, whether the differential treatment of women was an active choice.

3. Research Findings

The study reveals the motivations, behaviors, and real-life situations faced by women across three different stages of the "switch code" life cycle:

Motivations for Switching Code: Restrictions of the Educational System and the Desire for Economic Independence. The initial reasons women choose to switch code are deeply influenced by China's educational system and traditional gender concepts. The strict "arts and sciences division" system in high school, an exam-oriented education lacking career planning, and the traditional gender bias of parents and teachers who believe "liberal arts are more suitable for girls" have resulted in many women being excluded from IT and core STEM majors before entering university. Upon realizing the gender bias or salary ceilings in their original majors during college, they chose to learn programming. The rationale is that programming primarily relies on mental labor, where women face no physiological disadvantages, and the IT industry offers higher salaries and broader job opportunities, thereby helping them achieve economic independence and personal freedom.

Learning Process: Self-Study, Studying Abroad, and the Hindrance of "Benevolent Sexism". Due to strict restrictions on cross-major graduate entrance exams in mainland China, it is extremely difficult for women to directly transfer into computer-related master's programs domestically. Consequently, "switching code" is predominantly achieved through online self-study and studying abroad. During the learning process, the greatest technical difficulties they encounter are unrelated to gender, but rather stem from a lack of foundational knowledge. However, during campus group collaborations or academic research, they frequently encounter "benevolent sexism" from men. Male classmates often assign women repetitive tasks lacking technical substance under the guise of "taking care of women's emotions." This invisible exclusion essentially deprives women of the opportunity to receive core technical training. Fortunately, the mutually supportive and encouraging atmosphere of female communities on platforms like Douban provides them with robust emotional support to overcome learning difficulties.

Workplace Reality: Male Gaze, Marginalization, and the Compensation of High Salaries. Upon entering the IT workplace, women face a "chilly climate" that is even more severe than on campus. Social media (such as Zhihu) is rife with the male gaze and an excessive focus on female programmers' physical appearances. Meanwhile, the notion held by some men that women enjoy a "gender dividend" in IT job hunting is inconsistent with reality. In practice, the prevalent "996 overtime culture" in Chinese Internet companies and male-dominated "brotherhood" social circles result in women being severely marginalized in terms of promotions and team integration. Nevertheless, the relatively lucrative salaries and decent job opportunities in the IT industry mitigate the negative impacts of these gender challenges to a certain extent.

4. Conclusions and Policy Recommendations

Despite the existence of a chilly climate unfriendly to women in the IT industry, programming skills—as a tool for achieving higher income and career autonomy—propel Chinese women to continually break through barriers and proactively "switch code." Based on these findings,

this study recommends that Chinese high schools incorporate career awareness courses to prevent students from blindly choosing academic tracks due to gender stereotypes. Concurrently, universities should appropriately relax restrictions on undergraduate major transfers and provide more inclusive graduate programs for students lacking professional backgrounds.

Keywords: Internet Industry; Chinese Women; Programming; Women engineers; Gender bias

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